Imperium Annuity Agent — HOL Registry Broker Overview

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1) What Is the HOL Agent?

The **Imperium Annuity Agent** is a software agent registered on the **Hashgraph Online (HOL) Registry Broker** using the **HCS-10 OpenConvAI** standard. It is the on-chain identity that represents our annuity lifecycle engine to other agents and systems on the Hedera network.

In practical terms, the HOL Agent is:

- **A Hedera account** (`0.0.8196762`) with its own key pair, dedicated to agent operations.
- **A discoverable profile** inscribed on-chain via HCS-11, containing the agent's name, bio, capabilities, and skills.
- **A set of communication channels** (HCS topics) that enable other agents to discover, connect to, and exchange messages with our agent.
- **A registered entry** in the HOL Guarded Registry, making it searchable by any agent on the network.

The agent does not replace our CLI agent or API — it is the **identity layer** that wraps them, allowing the annuity lifecycle to be invoked by other agents over Hedera Consensus Service (HCS) rather than only through a local command line or REST calls.

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2) What Standards Does It Use?

Standard	Role	Description
HCS-10	Communication protocol	Defines how agents discover each other, establish connections, and exchange messages using HCS topics. The "OpenConvAI" standard.
HCS-11	Agent profile	Defines the agent's on-chain identity: display name, bio, capabilities, topic IDs, and metadata. Inscribed via HCS-1.
HCS-1	Data inscription	The underlying mechanism for storing data (like the agent profile JSON) on HCS topics.
HCS-2	Registry	Defines how agents are indexed in a shared registry topic for discovery.

All standards are maintained by the **Hashgraph Online DAO (HOL)** at `hol.org/docs/standards/` . They are not part of Hedera's core documentation — they are a community-driven layer built on top of Hedera Consensus Service.

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3) What Hedera Resources Were Created?

When we ran `node agent/hol-registry.js create` , the SDK created four Hedera resources — each a separate on-chain transaction:

Resource	Hedera ID	Purpose
Agent Account	`0.0.8196762`	The agent's own Hedera account with a dedicated key pair. Separate from our operator account (`0.0.7974882`). Funded with 50 testnet HBAR.
Inbound Topic	`0.0.8196678`	A public HCS topic where other agents send **connection requests**. Think of it as the agent's "inbox" for new connections.
Outbound Topic	`0.0.8196675`	A submit-key-gated HCS topic where the agent logs its own activity (connections accepted, messages sent). Only our agent can write to it — others can read it as a public audit trail.

Profile Topic	`0.0.8196782`	Stores the HCS-11 profile JSON (name, bio, skills, capabilities) inscribed via HCS-1. This is what other agents read to understand who we are and what we can do.

All four are visible on HashScan:

- Account: `https://hashscan.io/testnet/account/0.0.8196762`
- Inbound: `https://hashscan.io/testnet/topic/0.0.8196678`
- Outbound: `https://hashscan.io/testnet/topic/0.0.8196675`
- Profile: `https://hashscan.io/testnet/topic/0.0.8196782`

A fifth resource — the **Connection Topic** — is created dynamically each time two agents establish a connection. It uses a threshold key so both agents can write to it.

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4) How Was It Created and Registered?

4.1 The Creation Process

Our module `agent/hol-registry.js` uses the `@hashgraphonline/standards-sdk` (v0.1.165) to orchestrate the creation. The flow is:

```
AgentBuilder (define identity)
    ↓
HCS10Client.createAndRegisterAgent()
    ↓
Step 1: Create Hedera account (50 HBAR)
Step 2: Create outbound topic (submit-key-gated)
Step 3: Create inbound topic (public)
Step 4: Inscribe HCS-11 profile via HCS-1
Step 5: Register with HOL Guarded Registry
    ↓
State saved to deployments/hol-agent.json
```

4.2 The AgentBuilder Configuration

We used the SDK's fluent `AgentBuilder` API to define the agent's identity:

```
new AgentBuilder()
  .setName('Imperium Annuity Agent')
  .setAlias('imperium-annuity')
  .setBio('Australian Capital Markets annuity lifecycle agent by Imperium Markets...')
  .setType('autonomous')
  .setModel('cli-agent-v0.3')
  .setCreator('Imperium Markets')
  .setCapabilities([...])      // 7 AIAgentCapability enums
  .setNetwork('testnet')
  .addProperty('domain', 'Australian Capital Markets')
  .addProperty('skills', [...])  // 7 skill identifiers
  .addProperty('currency', 'AUD')
  .addProperty('dayCount', 'ACT/365')
  .addProperty('settlement', 'T+2')
```

4.3 The HCS10Client

The client connects to Hedera Testnet using our operator account's ECDSA key:

```
new HCS10Client({
  network: 'testnet',
  operatorId: '0.0.7974882',        // Our Hedera account
  operatorPrivateKey: '...',          // ECDSA hex key (same as Hardhat)
  keyType: 'ecdsa',
  guardedRegistryBaseUrl: 'https://moonscape.tech',  // HOL Registry
})
```

4.4 Resumable Creation

The creation process involves multiple Hedera transactions. If any step fails (e.g., the profile inscription timed out on our first attempt), the SDK saves a checkpoint. Running `create` again resumes from the last successful step — no resources are wasted or duplicated.

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5) What Is the Agent's On-Chain Profile?

The HCS-11 profile inscribed on topic `0.0.8196782` contains:

```
{
  "version": "1.0",
  "type": 1,
  "display_name": "Imperium Annuity Agent",
  "alias": "imperium_annuity_agent",
  "bio": "Australian Capital Markets annuity lifecycle agent by Imperium Markets. Issues,
settles, transfers, and redeems structured annuity products on Hedera. Supports ASIC compliance
checks, AUD formatting, ACT/365 day-count, and T+2 settlement.",
  "inboundTopicId": "0.0.8196678",
  "outboundTopicId": "0.0.8196675",
  "aiAgent": {
    "type": 1,
    "capabilities": [8, 10, 14, 18, 17, 7, 16],
    "model": "cli-agent-v0.3",
    "creator": "Imperium Markets"
  },
  "properties": {
    "domain": "Australian Capital Markets",
    "product": "Structured Annuities",
    "skills": [
      "annuity.issue", "annuity.settle", "annuity.transfer",
      "annuity.redeem", "annuity.compliance", "annuity.analytics",
      "annuity.audit"
    ],
    "version": "0.3.0",
    "currency": "AUD",
    "dayCount": "ACT/365",
    "settlement": "T+2"
  }
}
```

Capabilities (what the agent can do, generically)

Capability	Enum	Meaning
DATA_INTEGRATION	8	Read and integrate on-chain data

TRANSACTION_ANALYTICS	10	Analyse transaction patterns, yields, risk
COMPLIANCE_ANALYSIS	14	Regulatory compliance checks
WORKFLOW_AUTOMATION	18	Orchestrate multi-step processes
API_INTEGRATION	17	Interface with REST APIs
KNOWLEDGE_RETRIEVAL	7	Domain knowledge lookup
MULTI_AGENT_COORDINATION	16	Coordinate with other agents via HCS-10

Skills (what the agent can do, specifically)

Skill	Description
`annuity.issue`	Issue a new AnnuityToken with AusCM parameters
`annuity.settle`	Execute settlement (accept, fund, pay coupons)
`annuity.transfer`	Transfer annuity to secondary buyer
`annuity.redeem`	Redeem at maturity
`annuity.compliance`	Generate compliance report
`annuity.analytics`	Yield, duration, risk analytics
`annuity.audit`	Export full audit trail

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6) What Can We Do With It?

Today (Day 3 — implemented)

Action	Command	What happens
Create agent	`node agent/hol-registry.js create`	Creates Hedera account, topics, profile, registers with HOL
Check status	`node agent/hol-registry.js status`	Displays the agent's registered identity from `deployments/hol-agent.json`
Connect to another agent	`node agent/hol-registry.js connect <topic-id>`	Sends a connection request to another agent's inbound topic and waits for confirmation
Be discovered	Any agent calls `client.retrieveProfile('0.0.8196762')`	Returns our full profile with skills and capabilities

Tomorrow (Day 4 — planned)

Action	Description
Accept incoming connections	Monitor inbound topic, auto-accept connection requests, create Connection Topics
Receive skill invocations	Listen for messages on Connection Topics, parse skill requests (e.g., `annuity.issue`)

Execute skills via API	Map incoming HCS-10 messages to ImperiumAPI REST endpoints, execute on-chain
Respond with results	Send skill execution results back over the Connection Topic
Agent-to-agent demo	Create a second test agent, establish a connection, invoke a full annuity lifecycle

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7) How Does Agent-to-Agent Communication Work?

The HCS-10 connection flow follows a 4-step handshake:

```
Agent A (requester)                          Agent B (our agent)
        |                                          |
        |  1. connection_request                   |
        |──────────────────────────────────────────→ |  (on B's Inbound Topic)
        |                                          |
        |  2. connection_created                   |
        | ←────────────────────────────────────────── |  (B creates Connection Topic)
        |                                          |
        |  3. message (skill invocation)           |
        |──────────────────────────────────────────→ |  (on Connection Topic)
        |                                          |
        |  4. message (skill result)               |
        | ←────────────────────────────────────────── |  (on Connection Topic)
        |                                          |
```

1. **Agent A** discovers our agent in the HOL Registry and sends a `connection_request` to our **Inbound Topic** (`0.0.8196678`).
2. **Our agent** monitors the inbound topic, sees the request, creates a new **Connection Topic** (threshold key — both agents can write), and sends a `connection_created` response.
3. **Agent A** sends a `message` on the Connection Topic containing a skill invocation (e.g., `{"skill": "annuity.issue", "params": {...}}`).
4. **Our agent** processes the request, executes the on-chain operation via ImperiumAPI, and sends the result back on the same Connection Topic.

All messages are recorded on Hedera Consensus Service — immutable, timestamped, and auditable.

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8) How Does This Differ From Hedera's Free Tools?

Aspect	Hedera Portal (Contract Builder / Playground)	Imperium HOL Agent
Identity	No agent identity — user interacts directly	On-chain agent identity with HCS-11 profile, discoverable by other agents
Communication	None — single-user tool	HCS-10 agent-to-agent protocol, multi-party workflows
Domain knowledge	Generic contract interaction	Australian Capital Markets: ACT/365, T+2, AUD, ASIC compliance
Skills	None — manual contract calls	7 published skills with structured invocation and response
Audit trail	Basic transaction explorer	Structured HCS message log, exportable compliance reports

Automation	Manual	Autonomous agent type, automated skill execution

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9) Architecture Diagram

```
┌──────────────────────────────────────────────────────────────┐
│                    Hedera Testnet                            │
│                                                              │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────────┐ │
│  │ Inbound      │  │ Outbound     │  │ Connection Topics    │ │
│  │ Topic        │  │ Topic        │  │ (created per conn)   │ │
│  │ 0.0.8196678  │  │ 0.0.8196675  │  │                      │ │
│  └──────┬───────┘  └──────┬───────┘  └──────────┬───────────┘ │
│         │                 │                     │             │
│         │                 │                     │             │
│  ┌──────┴─────────────────┴─────────────────────┴──────────┐ │
│  │             Agent Account  0.0.8196762                   │ │
│  │             Profile Topic  0.0.8196782                   │ │
│  └──────────────────────────┬──────────────────────────────┘ │
│                             │                                │
│                             │                                │
│  ┌──────────────────────────┴──────────────────────────────┐ │
│  │        HOL Guarded Registry (moonscape.tech)             │ │
│  │        Agent discoverable by name, skills, capabilities  │ │
│  └──────────────────────────────────────────────────────────┘ │
│                                                              │
│  ┌──────────────────────────────────────────────────────────┐ │
│  │ ImperiumStableCoin  0xC44f...Af13                        │ │
│  │ AnnuityToken        0x6e5A...99d0                        │ │
│  │ (EVM smart contracts on Hedera via Hashio JSON-RPC)      │ │
│  └──────────────────────────────────────────────────────────┘ │
└──────────────────────────────────────────────────────────────┘
                             │
                             │ JSON-RPC (Hashio)
                             ▼
┌──────────────────────────────────────────────────────────────┐
│                    Imperium Infrastructure                   │
│                                                              │
│  ┌──────────────────┐    ┌──────────────────────────────┐   │
│  │ agent/           │    │ api/                         │   │
│  │ hol-registry.js  │───→│ imperium-api.js              │   │
│  │ cli-agent.js     │    │ (10 REST endpoints)          │   │
│  └──────────────────┘    └──────────────────────────────┘   │
│                                                              │
│  ┌──────────────────────────────────────────────────────────┐ │
│  │ deployments/hol-agent.json  (agent state)                │ │
│  │ deployments/hedera-testnet.json  (contract addresses)    │ │
│  └──────────────────────────────────────────────────────────┘ │
└──────────────────────────────────────────────────────────────┘
```

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10) Key Files

File	Purpose
`agent/hol-registry.js`	CLI module for agent creation, status, and connection. Uses `@hashgraphonline/standards-sdk`.

`deployments/hol-agent.json`	Persisted agent state: account ID, topic IDs, private key, skills.
`.env`	Operator credentials (`HEDERA_TESTNET_ACCOUNT_ID`, `HEDERA_TESTNET_PRIVATE_KEY`).
`agent/cli-agent.js`	The interactive CLI agent (v0.3) — currently operates via REST; Day 4 will connect it to HCS-10 messaging.
`api/imperium-api.js`	The ImperiumAPI — 10 REST endpoints that execute on-chain operations. The HOL agent will invoke these when processing skill requests from other agents.

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11) Costs

All operations on Hedera Testnet use free testnet HBAR. On mainnet, the costs would be:

Operation	Estimated Cost
Agent creation (account + 2 topics + profile inscription + registration)	~5-10 HBAR
Each HCS message (connection request, message, close)	~0.0001 HBAR
Smart contract execution (via Hashio JSON-RPC)	Variable (gas-based)

HCS messages are extremely cheap — fractions of a cent per message at current HBAR prices.

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12) Next Steps (Day 4)

1. **Inbound message handler** — Poll the inbound topic for `connection_request` operations, auto-accept, and create Connection Topics.
2. **Skill execution bridge** — Map incoming HCS-10 skill invocations to ImperiumAPI REST endpoints.
3. **Agent-to-agent test** — Create a second "investor" agent, connect to our agent, and execute a full annuity lifecycle over HCS-10.
4. **Integrate with CLI agent** — Allow the CLI agent to operate in "HCS-10 mode" where it receives commands from other agents rather than stdin.